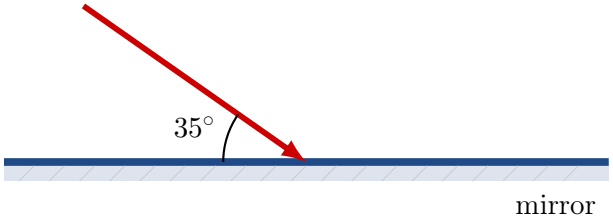
1. A laser beam hits a flat mirror. The beam makes an angle of 35° with the **surface** of the mirror, as shown.

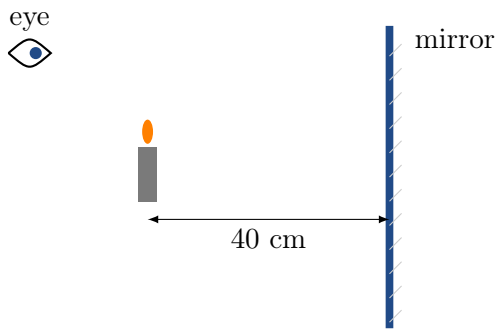


A. On the figure, draw the **normal** and the **reflected** beam. Label θ_i and θ_r.

B. What is the angle of incidence θ_i?

C. What is the angle between the incident beam and the reflected beam?

2. A candle stands 40 cm in front of a flat mirror. Your eye is off to the side.

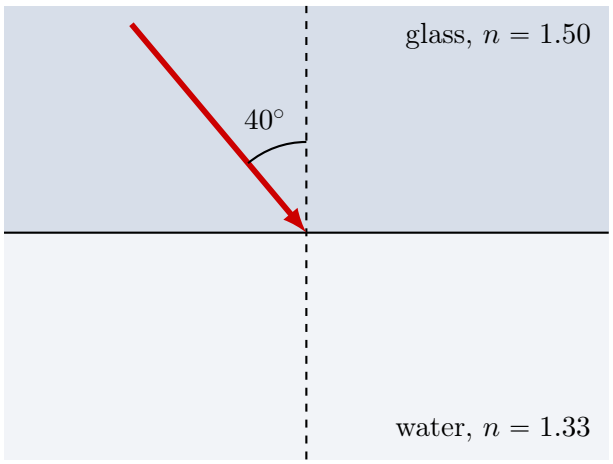


A. Using the law of reflection, draw **two** rays that leave the candle flame, bounce off the mirror, and reach the eye.

B. Extend the reflected rays **behind** the mirror (dashed) and mark where the image is. How far behind the mirror is it?

C. Is the image real or virtual? Explain using the words "rays go" or "rays appear to go."

3. A ray of light travels from glass ($n = 1.50$) into water ($n = 1.33$). It hits the boundary at 40° from the normal.



A. Find the refracted angle θ_2.

B. Does the ray bend **toward** or **away from** the normal? How can you tell without a calculator?

C. Draw the refracted ray on the figure. Also draw the weak reflected ray.